

III. EDA-Data Cleaning

AIM:

- Handling missing values: detection, filling, and dropping
- Removing duplicates and unnecessary data
- Data type conversion and ensuring consistency
- Normalize data (e.g., standardization, min-max scaling).

PROGRAM AND OUTPUT:

```
# Importing required libraries
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.preprocessing import MinMaxScaler, StandardScaler

# Load the dataset
df = pd.read_csv('/content/test_Y3wMUE5_7gLdaTN.csv')

# Display basic information
print("Initial Data Overview:")
print(df.info())
```

```
Initial Data Overview:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 367 entries, 0 to 366
Data columns (total 12 columns):
#   Column              Non-Null Count  Dtype
---  -
0   Loan_ID              367 non-null   object
1   Gender               356 non-null   object
2   Married              367 non-null   object
3   Dependents           357 non-null   object
4   Education            367 non-null   object
5   Self_Employed        344 non-null   object
6   ApplicantIncome      367 non-null   int64
7   CoapplicantIncome    367 non-null   int64
8   LoanAmount           362 non-null   float64
9   Loan_Amount_Term     361 non-null   float64
10  Credit_History       338 non-null   float64
11  Property_Area        367 non-null   object
dtypes: float64(3), int64(2), object(7)
memory usage: 34.5+ KB
None
```

1. Handling Missing Values

```
print("\nMissing Values in Each Column:\n", df.isnull().sum())

sns.heatmap(df.isnull(), cbar=False, cmap="Blues")

plt.title("Missing Value Heatmap")

plt.show()

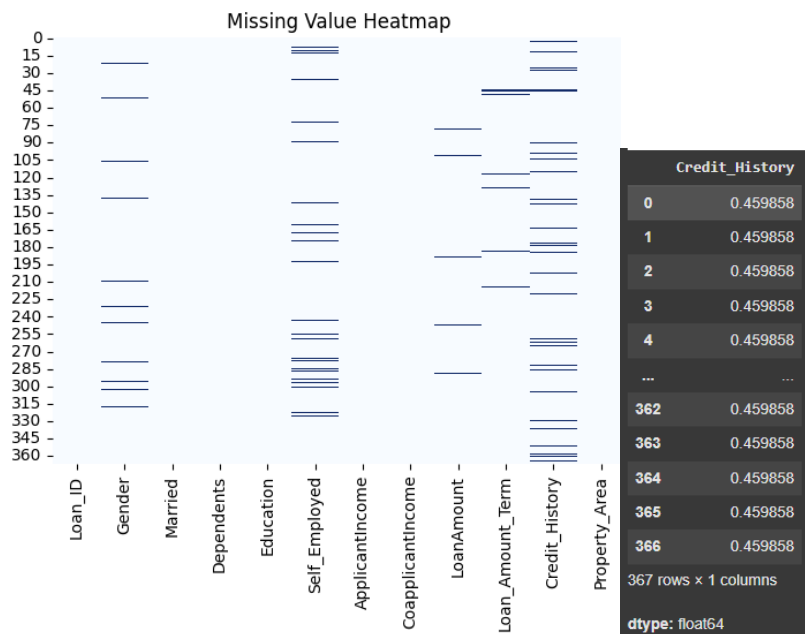
for col in ['Gender', 'Married', 'Dependents', 'Self_Employed']:
    df[col].fillna(df[col].mode()[0])

df['LoanAmount'].fillna(df['LoanAmount'].median())

df['Loan_Amount_Term'].fillna(df['Loan_Amount_Term'].mode()[0])

df['Credit_History'].fillna(df['Credit_History'].mode()[0])
```

```
Missing Values in Each Column:
Loan_ID              0
Gender               11
Married              0
Dependents           0
Education            0
Self_Employed       23
ApplicantIncome      0
CoapplicantIncome    0
LoanAmount           5
Loan_Amount_Term     6
Credit_History       29
Property_Area        0
dtype: int64
```



2. Removing Duplicates

```
initial_rows = df.shape[0]
df.drop_duplicates(inplace=True)
print(f"\nRemoved {initial_rows - df.shape[0]} duplicate rows.")
```

```
Removed 0 duplicate rows.
```

3. Data Type Conversion

```
# Convert 'Dependents' to numeric (replace '3+' with 3)
df['Dependents'] = df['Dependents'].replace('3+', 3).fillna(0).astype(int)
```

4. Ensuring Categorical Consistency

```
for col in ['Gender', 'Married', 'Education', 'Self_Employed', 'Property_Area']:
    df[col] = df[col].str.strip().str.capitalize()
```

5. Normalization

```
min_max_scaler = MinMaxScaler()
scale_cols = ['ApplicantIncome', 'CoapplicantIncome', 'LoanAmount']
```

```

df[scale_cols] = min_max_scaler.fit_transform(df[scale_cols])

scaler = StandardScaler()

df[['Credit_History']] = scaler.fit_transform(df[['Credit_History']])

# 6. Final Overview

print("\nCleaned Data Summary:")

print(df.info())

print(df.head())

```

```

Cleaned Data Summary:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 367 entries, 0 to 366
Data columns (total 12 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Loan_ID                367 non-null   object
1   Gender                 356 non-null   object
2   Married                367 non-null   object
3   Dependents             367 non-null   int64
4   Education              367 non-null   object
5   Self_Employed          344 non-null   object
6   ApplicantIncome        367 non-null   float64
7   CoapplicantIncome      367 non-null   float64
8   LoanAmount             362 non-null   float64
9   Loan_Amount_Term       361 non-null   float64
10  Credit_History         338 non-null   float64
11  Property_Area          367 non-null   object
dtypes: float64(5), int64(1), object(6)
memory usage: 34.5+ KB
None
   Loan_ID  Gender  Married  Dependents  Education  Self_Employed  \
0  LP001015  Male     Yes         0      Graduate             No
1  LP001022  Male     Yes         1      Graduate             No
2  LP001031  Male     Yes         2      Graduate             No
3  LP001035  Male     Yes         2      Graduate             No
4  LP001051  Male     No          0  Not graduate             No

   ApplicantIncome  CoapplicantIncome  LoanAmount  Loan_Amount_Term  \
0      0.078865      0.000000      0.157088      360.0
1      0.042411      0.062500      0.187739      360.0
2      0.068938      0.075000      0.344828      360.0
3      0.032263      0.106083      0.137931      360.0
4      0.045168      0.000000      0.095785      360.0

   Credit_History  Property_Area
0      0.459858      Urban
1      0.459858      Urban
2      0.459858      Urban
3      NaN         Urban
4      0.459858      Urban

```

RESULT:

Thus, the program was written and executed successfully.